Feasible Reform of a Persistent Distortion: The Political Economy of Coal Allocation in India*

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Abstract

India allocates coal to power plants at notified prices through rationed supply contracts that favour state-owned generators, so the cleanest and most efficient generators pay the most for their fuel. The arrangement has survived governments of every political stripe. I argue that it persists because Coal India Ltd (CIL) is a common agent: it answers to several principals with conflicting claims, including the fiscal authority, the coal-bearing states, the cheap-power constituency and the states that own generators, and the result is governance by prohibition rather than by incentive. The state solved the commitment problem for private investors in electricity but never in coal, and reforms that ignored a principal's claim were reversed. A feasible reform must leave each principal no worse off on the dimension it controls, either by compensating principals within the existing prohibitions or by giving each its own instrument. I evaluate one reform of each kind, both with precedents in Indian policy. A compensatory tariff and a deregulation of coal prices paired with direct benefit transfers to consumers produce the same reallocation of generation, a $4.7 billion annual welfare gain and a 14.6 million tonne annual reduction in CO_2 emissions. The first costs the central budget $4.9 billion a year and lowers electricity bills by $7.3 billion; in the second, CIL's higher revenue funds the transfers and the fuel and transport savings remain with generators. In both, state-owned generators lose a fifth of their output, worth at most 0.38 percent of any state's GDP.

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1 Introduction

Economists often judge an economic outcome by whether a feasible superior alternative can be described and implemented with net gains, where feasibility is set by the constraints of resources and technology. In developing economies, this standard has taken the form of reforms that aim to get prices right by letting market forces determine prices, opening entire sectors to private entry and privatizing state-owned enterprises (Timmer, 1986; Williamson, 1990; Megginson and Netter, 2001). A large literature explains why such reforms are so often blocked, diluted or reversed: lobbying from special interest groups (Grossman and Helpman, 1994), uncertainty about the benefits from reform (Fernández and Rodrik, 1991), and how the beneficiaries of inefficient policies help keep them in place (Coate and Morris, 1999). Since reforms that improve efficiency might not always be feasible or come with large fixed costs, one response is to change the policy prescription rather than the politics: instead of building the formal institutions that first-best reforms require, work with second-best institutions that fit local constraints ((Qian, 2003); (Rodrik, 2008)). However, empirical assessments of feasibility are rare in this literature: few studies identify the groups able to block a reform, design the transfers needed to secure their consent, and measure the resulting efficiency gains and fiscal costs.

This paper studies one such inefficient policy, the allocation of coal to power plants in India, and the political process that sustains it. India's coal allocation is a case in which the efficiency cost is large and the textbook solution is well known. Coal India Limited (CIL), a state-owned near-monopoly, sells coal at administered prices well below the market benchmarks and rations it through long-term supply contracts, the priority for which goes to inefficient state-owned generators. Generators without enough contracted coal, most of them private, buy the rest at market prices. As a result, the cleanest and most efficient generators face the highest fuel costs. An alternative policy that prices coal uniformly across generators would save $1.9 billion a year in energy and transport costs and cut CO_2 emissions by 14.6 million tonnes (Deshwal, 2026). The textbook solution– deregulate coal prices and privatize mining is not politically feasible (Reddy and Sumithra (1997)). Public monopoly over coal has survived successive governments with very different commitments to markets, including those that privatized electricity generation. This paper explains the persistence of the arrangement and identifies reforms that improve on the status quo within the existing political process.

Methodologically, I treat the political process as a constraint that can be described and measured. Following Dixit (1996), I add the transaction costs of the political process to the resource and technology constraints that define the feasible set of economic policies. Williamson (1996)'s remediableness test then requires that a superior alternative to the current coal policy be implementable through that process with net gains. I identify the principals able to block a change and the margin each controls, construct reforms that leave each principal no worse off on that margin, and quantify the efficiency gain, its distribution and the residual losses. The reforms are approximately Pareto improving: no principal with veto power loses on the margin it controls.

I argue that the allocation persists because CIL is a common agent. It answers to several principals with conflicting claims: the fiscal authority (central government), the coal-bearing states, the cheap-power constituency, the states that own generators, generating companies with and without supply contracts, and electricity consumers at large. A principal that rewards CIL for the actions it values finds part of the reward leaking to principals with opposing interests. Because of this free-rider problem, every principal offers weak incentives in the Nash equilibrium, and CIL's actions are therefore better influenced by prohibiting some of the actions than by rewarding others (Holmstrom and Milgrom, 1991; Dixit, 1996). The result is governance by prohibition: limits on CIL's prices, customers, contract terms and exit, layered so that any change on any margin requires the assent of every principal able to retaliate ((Wilson, 1989)) The argument has three parts. First, the Indian state solved the commitment problem for private investors in electricity but not in coal, so private generators entered on credible terms in the product market and on revocable terms in the input market. Second, the common-agency structure of CIL's governance explains why the input market was never reformed and why reforms that ignored any principal were reversed. Third, a reform is feasible only if it leaves each principal that can block it no worse off on the dimension it controls. Two designs meet this condition: compensate the principals within the existing prohibitions, or compartmentalise, giving each principal its own instrument so that the mutual free-riding disappears and no principal needs the coal price to pursue its objective (Dixit, 1996).

I then evaluate one reform of each kind, both with precedents in Indian policy, using the model of India's electricity supply from the companion paper (Deshwal, 2026). A *compensatory tariff* compensates within the prohibitions. It keeps notified prices,

supply contracts and allocation rules, and the central government pays private generators the gap between the market and notified price on the coal they buy beyond their contracts. *Price deregulation with a direct benefit transfer (DBT)* compartmentalises. It prices all coal at the market price and meets the cheap-power objective with a separate instrument, a cash transfer that pays each state's consumers the increase in their electricity bills. CIL remains a public enterprise, with its output, workforce and royalty payments unchanged. Neither reform needs a commitment device outside the existing political process. Under both, every principal that can block the reform retains the levers of influence it holds today and is no worse off under the reforms. If the Centre stopped paying the compensation or the transfer, the principals it harmed could retaliate through these levers, for example by withdrawing support from the governing coalition, so the Centre gains nothing by reneging.

Both reforms give every generator the same price of coal at the margin, so they produce the same reallocation of generation and the same annual gain of $4.7 billion: $1.9 billion in fuel and transport savings and $2.8 billion in avoided CO_2 damages from 14.6 million fewer tonnes of emissions. They differ in the distribution of the gain. Under the compensatory tariff, the average contract price of electricity falls from INR 3.30 to 2.94 per kWh and the average short-term price from INR 5.26 to 4.09 per kWh. Consumers' bills fall by $7.3 billion a year, and the Centre pays $4.9 billion net of CIL's gains. Under deregulation, the average contract price rises to INR 6.50 per kWh and bills rise by $47.2 billion a year. CIL's additional revenue of $47.1 billion funds a transfer that holds the welfare of every state's consumers unchanged, so neither the consumers nor the Centre is worse off. Despite these differences, the compensatory tariff is not necessarily the more feasible of the two. The government decides the size of the transfer: to win consumers' support against the states whose generators lose, it could direct to consumers an amount equal to their gain under the compensatory tariff, and the two reforms would then be, on average, equally feasible.

In both reforms, the only principal that loses is the state government that owns generators: state-owned generation falls by 72 TWh a year, a fifth of its status-quo level. The impacts of the output loss are primarily revenue and employment effects from lower utilization of generators, and the political value that state governments attach to their own generators. The economic effects are small relative to state economies: valued at the average status-quo price of coal and lignite generation, the lost output is worth at most 0.38 percent of a state's GDP, and 0.12 percent

in the median state. A state government weighs this loss against the gain to its consumers. Under the compensatory tariff, consumers' gain exceeds the value of the lost output in 13 of the 14 states with state-owned generators, and a state government obstructs the reform only if it values a dollar of its generators' output well above a dollar of its consumers' gains (Grossman and Helpman, 1994; Shleifer and Vishny, 1994). Moreover, coal allocation is decided by the Centre, and the Centre does not necessarily depend on the support of every state that owns generators–many of these states may not be pivotal to the governing coalition at any given time. The reforms therefore pass the remediableness test with a qualification on the objective of these state governments, which I argue is a minor obstacle.

In terms of related literature, the paper contributes primarily to the literature on the political economy of reforms, and in particular to the transaction-cost politics literature (Dixit, 1996; Williamson, 1996). The framework is usually applied to the evaluation of economic policies only descriptively. In this paper, I measure the costs and benefits of feasible alternatives to the status quo for each principal able to block them empirically, so that the *remediableness* test is carried out rather than only described. I show that the efficiency cost of a subsidy can lie in its design, not its existence, so that the different political claims that sustain it can be honored while the distortionary feature of the design is removed. The paper also adds to work on the political economy of electricity subsidies in India (Dubash and Rajan, 2001; Burgess et al., 2020; Montrone et al., 2022), which has focused on retail tariffs and distribution, by locating a major source of inefficiency upstream, in the governance of coal.

The rest of the paper proceeds as follows. Section 2 describes the distortion and its cost. Section 3 sets out the framework and the evidence for each part of the argument. Section 4 describes the two reforms and their precedents, Section 5 explains how I evaluate them, and Section 6 reports the distribution of impacts across principals and states and its implications for the remediableness test. Section 7 concludes.

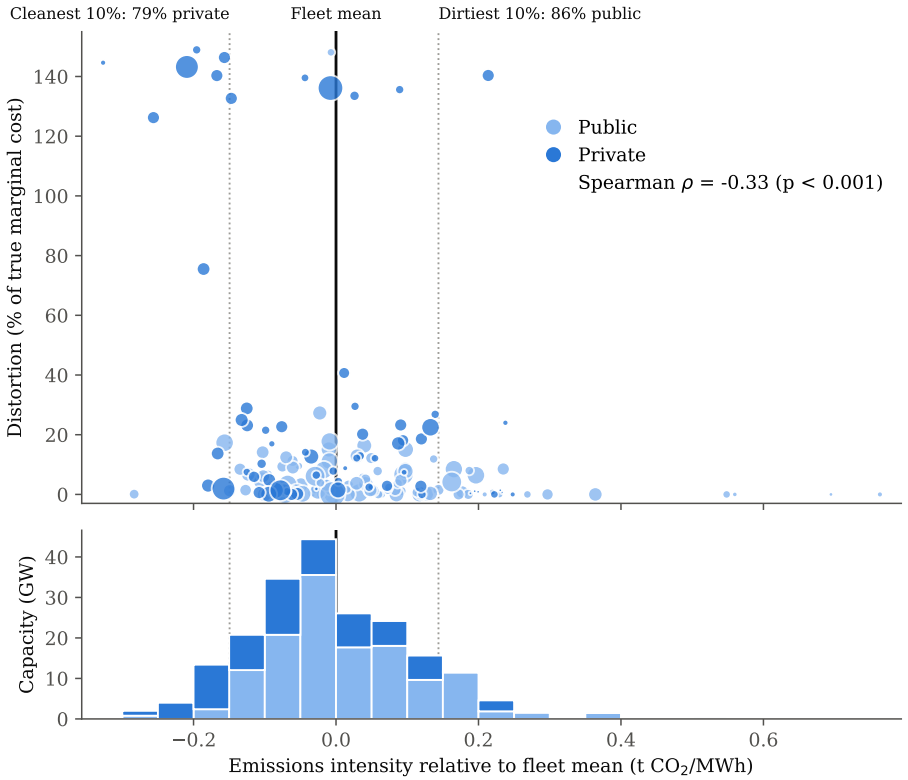
2 The Distortion and Its Cost

Coal generates about three quarters of India's electricity, and CIL produces about 84 percent of India's domestic coal. Most of that coal is sold to power plants at notified prices, administratively determined, grade-specific prices well below the import-parity price, under long-term supply contracts, known as fuel supply agreements. Access to

these contracts is rationed, and allocation rules give priority to public-sector generators. Generators without enough contracted coal buy the rest through e-auctions or imports, at prices up to two and a half times higher per unit of energy (Deshwal, 2026).

The distortion that matters for efficiency is the difference in the price generators pay for their *marginal* tonne of coal. A state-owned generator with a supply contract burns an additional tonne at the notified price; a private generator without one buys it at the market price. Because private generators are newer and more efficient, the generators with the lowest true costs and emissions face the highest private costs. In the companion paper, I measure this distortion directly: it averages 5 percent of true marginal cost for public generators and 43 percent for private generators (Deshwal, 2026). Figure 1 plots each generator's distortion against its emissions intensity. The two are negatively correlated. The cleanest tenth of capacity is 79 percent private and faces an average distortion of nearly two-thirds of its true marginal cost; the dirtiest tenth is 86 percent public and faces a distortion five times smaller.

Figure 1. The distortion, by ownership and emissions intensity



The distortion is the difference between a generator's observed marginal cost and its marginal cost when all of its coal is priced at the CIL or SCCL notified price per kcal, holding coal quality, taxes and transport costs fixed, expressed as a percentage of this undistorted (true) marginal cost. Top: each marker is one of 172 coal and lignite generators, with area proportional to installed capacity; emissions intensity is measured relative to the capacity-weighted fleet mean of 1.013 tonnes of CO_2 per MWh. The solid vertical line marks the fleet mean, and the dotted lines the capacity-weighted 10th and 90th percentiles of emissions intensity; the labels report the private and public shares of capacity in the cleanest and dirtiest tenths. Bottom: installed capacity by emissions intensity, in bins of 0.05 tonnes of CO_2 per MWh, stacked by ownership. Averages over July 2021–December 2023. Source: Deshwal (2026).

Removing the distortion, by giving every generator the same price of coal per kcal at the margin, reallocates generation from state-owned to private generators. Private generation rises by 24 percent and state-owned generation falls by 21 percent, while central-sector generation falls by only 4 percent. The reallocation saves $1.9 billion a year in fuel and transport costs and reduces CO_2 emissions by 14.6 million tonnes, more than a $40 per tonne carbon tax would achieve. Valuing CO_2 at $190 per tonne, the annual gain is $4.7 billion.

Removing the distortion yields large gains, and they accrue to several parties. Private generators produce more. The savings in fuel and transport costs can be shared

between electricity consumers, through lower prices, and the fiscal authority and CIL, through the revenue from coal that state-owned generators no longer burn. Lower emissions of CO_2 and local air pollutants benefit consumers and society at large. The losers are primarily state-owned generators, whose lost output is small relative to the economies of the states that own them (Section 6). In a political process without transaction costs, the winners would compensate the losers and the distortion would be removed. Its persistence across two decades of reform therefore points to the costs of political transactions, for example, the difficulty of making credible commitments to private investors and operators in the coal market, which would allow coal to be competitively priced, and the difficulty of securing the agreement of the many principals with claims on CIL. The next section develops this argument.

3 The Persistence of the Distortion: A Transaction-Cost Politics View

The political process of economic policy matters, and understanding it matters for how we judge and evaluate a policy. A test of the inefficiency of an economic outcome must recognise the constraints imposed by transaction costs in the political process just as it respects resource and technology constraints (Dixit, 1996). Williamson (1996) formalizes this as a test of *remediableness*: an outcome is presumed efficient unless a feasible superior alternative can be described and implemented with net gains, where feasibility is judged against the transaction technology of the political process itself. This section applies the framework to India's coal allocation. I first describe two institutional features that give CIL's principals their claims, and the principals themselves. I then develop two arguments. The first concerns commitment: the Indian state built devices that made its promises to private investors credible in electricity, but not in coal. The second concerns common agency: because CIL serves several principals with conflicting claims, it is governed by prohibitions rather than incentives, and no principal can delegate the coal market to an independent authority without the consent of the others. Together, the two arguments account for the persistence of the distortion across governments of different parties, and they determine the form of a feasible reform, the subject of Section 4.

3.1 Two institutional features

Two features of India's political economy give the principals of CIL claims that the political process treats as binding. Both predate the reforms of the 1990s and 2000s that opened electricity generation to private investment and changed the rules for allocating coal, and both have constrained the form those reforms took.

The first is what may be called the *public-interest mandate*. The provision of sufficient and affordable electricity has, since independence, been treated by successive Indian governments as a major objective of national policy in its own right (Montrone et al., 2022). Cheap, domestically abundant coal has been seen as the natural instrument for meeting it, and the structure of the sector has been shaped accordingly: the dominant role of the public sector in generation, transmission and distribution, the extension of grid access to rural areas, and the attention paid to tariffs for poorer consumers. The mandate has been reinforced by the politics of electricity subsidies (Dubash and Rajan, 2001). The expansion of groundwater irrigation during the Green Revolution turned farmers into a politically organised consumer class, and from the late 1970s, beginning with the flat-rate tariff offered to farmers in the 1977 Andhra Pradesh election, state-level parties came to treat agricultural power subsidies as a routine instrument of electoral competition. Adopted in turn by Tamil Nadu, Karnataka, Maharashtra and Punjab, among others, these subsidies gave cheap electricity a constituency of its own and made any reform that visibly raises tariffs politically costly (Dubash and Rajan, 2001). More recently, the state has come to treat electricity access as a right, a view that has driven universal electrification programmes (Burgess et al., 2020). The mandate sits uneasily with a conception of the sector as a commercial enterprise in which prices reflect costs and investment is determined by competition.

The second feature is the position CIL has come to occupy. CIL is not a coal producer that happens to be owned by the state. Chandra (2018) argues that it has established itself as a development actor in its own right: sustaining employment at a scale few other entities in India can match, anchoring infrastructure and welfare provision in the regions where it operates, and contributing to state finances through royalties. The government has used the corporation to foster investment, create employment and redistribute resources in the coal-rich regions, and state-level parties whose support is necessary to coalitions at the Centre have pressed for continued investment in CIL's operations in their constituencies (Montrone et al., 2022). The view of state-owned enterprises as drags on productivity has had little purchase in the political

discourse surrounding CIL (Chandra, 2018). CIL has remained, and has been treated as something that must remain.

3.2 The principals of CIL

CIL serves several principals, listed in Table 1. Each can influence CIL's actions through one or more levers: the Centre's ownership of CIL and its control of the Ministry of Coal, support for coalitions at the Centre, Centre–state relations, elections, lobbying and political funding, and litigation in the courts. Their interests in CIL's output conflict.

The fiscal authority looks to CIL's dividends and would like it to operate efficiently and profitably, since higher revenue keeps budget deficits in check. The coal-bearing states draw royalties from CIL and political legitimacy from the employment its operations sustain, and would like it to produce more; many are governed by parties whose support is necessary to coalitions at the Centre. The cheap-power constituency, mainly agricultural consumers organised by state-level parties around subsidised tariffs, has a powerful if indirect interest in the price CIL charges thermal generators, and acts through elections. State governments that own generators look to them for revenue, public employment and a reliable supply of electricity, expect priority in coal allocation, particularly during shortages, and press these claims through Centre–state relations and coalition politics. Generating companies with supply contracts benefit from coal at notified prices; the large firms among them also benefit from subsidised coal for non-power uses such as steel and cement, and some maintain close ties with the central government and contribute to the financing of its campaigns (Montrone et al., 2022). Generating companies without supply contracts buy their marginal coal at market prices and would benefit from competitive coal pricing, but their influence is limited to litigation and to lobbying with little weight.

Electricity consumers at large, the domestic, commercial and industrial consumers outside the cheap-power constituency, also have a stake in CIL's allocation. Retail tariffs in India are cross-subsidised: commercial and industrial consumers pay tariffs above the average cost of supply, while domestic and agricultural consumers pay subsidised tariffs below it (Central Electricity Authority, 2024). Holding these cross-subsidies fixed, every class of consumer pays more than it would under an efficient allocation of coal, because the allocation raises the cost of the generation that serves them. Consumers also bear the environmental costs of the allocation: by favouring

older and less efficient generators, it raises emissions of CO_2 and of local air pollutants. Their interests are diffuse, and the vote is their most important lever.

Table 1. The principals of CIL

Principal	Objective	Lever of influence	What the status quo gives it
Fiscal authority (the Centre)	Dividends, fiscal balance	Ownership; Ministry of Coal	CIL dividends; higher revenue keeps budget deficits in check
Coal-bearing states	Royalties, regional employment	Support for coalitions at the Centre	Royalties; CIL employment and investment
Cheap-power constituency	Low electricity tariffs	Elections	Cheap coal, hence cheap thermal power
States that own generators	Revenue, public employment, reliable electricity supply	Centre–state relations; support for coalitions at the Centre	Priority access to coal at notified prices
Generating companies with supply contracts	Rents on contracted coal	Lobbying; political funding	Contracted coal at notified prices, below market prices
Generating companies without supply contracts	Competitive coal pricing	Courts; lobbying, with limited weight	Marginal coal at market prices
Electricity consumers at large	Affordable, reliable electricity; clean air	Elections	Inefficient electricity prices, lower consumer surplus and higher emissions

The cheap-power constituency comprises consumers, mainly agricultural, whose tariffs are subsidised and whom state-level parties organise around those subsidies. Electricity consumers at large are domestic, commercial and industrial consumers. Notified prices are CIL's administratively set prices for coal sold under supply contracts; market prices are the prices of coal bought through e-auctions or imports. Sources: Dubash and Rajan (2001), Chandra (2018), Montrone et al. (2022).

3.3 Commitment and delegation in two markets

The commitment problem. A firm invests when the expected return on an investment exceeds its cost. In electricity generation, investment is large, long-lived and specific to its use, so once the capital is sunk, the return depends on later decisions of the state: the prices it allows, the inputs it makes available and the enforcement of contracts. Political opportunism makes these returns uncertain, especially where contracts are weakly enforced, and because investment is irreversible, the uncertainty

lowers investment in equilibrium (Dixit and Pindyck, 1994). Both risks are present in Indian electricity: weakly enforced contracts are renegotiated (Ryan, 2020), and generators that sell to state-owned distribution companies face the risk of hold-up. When commitment is weak, investment is therefore lower, all else equal, or investors demand a risk premium ((Ryan, 2021)). Levy and Spiller (1994) show that utility sectors are especially exposed to this problem, because they combine large specific investments, economies of scale and widespread consumption, so that prices are politically salient and the gains from opportunism are large. Electricity has all three features, and so does the supply of coal to power plants, for which most generators depend on a single seller.

Delegation as a commitment device. Governments address the commitment problem by delegating authority to a body that does not share their temptation to renege, either because its preferences differ from those of the government or because its mandate is narrow and specified in advance (Rogoff, 1985). Delegation to such non-majoritarian institutions also lets governments draw on expertise and shift the blame for unpopular decisions (Thatcher and Stone Sweet, 2002). Delegation does not remove politics from the sector; it raises the cost of reneging, because a reversal then requires changing a statute, breaking a contract or overruling an adjudicator rather than issuing a directive. Levy and Spiller (1994) identify three layers of a credible regulatory framework: substantive restraints on the discretion of the regulator, such as rules written into licences and contracts; restraints on changing the regulatory system itself, such as statutes that are costly to amend; and institutions that enforce both, such as an independent judiciary. The credibility of the framework depends on all three.

Electricity. The first wave of private participation in India's power sector, opened by the Electricity Laws (Amendment) Act of 1991, lacked all three layers and largely failed. Potential entrants faced monopsony buyers in the vertically integrated State Electricity Boards and, without deeper structural reform, were unwilling to commit capital on the strength of policy promises (Mathavan, 2008). The reforms that followed built each layer. The Electricity Regulatory Commissions Act of 1998 and the Electricity Act of 2003 created independent regulatory commissions at the Centre and in each state, established by statute and therefore costly to dismantle. The Electricity Act and the National Tariff Policy of 2006 required state procurement of power through competitive bidding, so that the price of a new generator's output was

set by its own bid and written into a long-term power purchase agreement, typically for 25 years, with a capacity charge that recovers fixed costs on available capacity and an energy charge that covers fuel. The commissions approve these contracts and adjudicate disputes between generators and buyers, with appeal to the Appellate Tribunal for Electricity and the Supreme Court. The State Electricity Boards were unbundled into separate generation, transmission and distribution entities. By insulating price setting, contracts and their adjudication from political interference, these reforms reduced the risk of expropriation in the product market. By 2019, private generators held over 36 percent of capacity, more than either the states or the central government.

Coal. None of the three layers was built for coal. Since the Coal Mines (Nationalisation) Act of 1973, which transferred coal holdings to CIL, private generators had little access to domestic coal. After private companies took the matter of access to the Supreme Court, the New Coal Distribution Policy of 2007 (Ministry of Coal, 2007) introduced Letters of Assurance leading to fuel supply agreements that promised full supply to existing and new generators, public and private. These promises were policy, not statute, and government directives later revised them. A Presidential directive of 2012 required CIL to supply 100 percent of the coal needed by 78 GW of capacity planned for 2012–17; a revision in 2013 reduced this to 60 GW.[1] The Ministry of Coal then set differentiated trigger levels, the minimum supply below which CIL pays a penalty, of 90 percent for plants contracted before the 2007 policy, mostly public, and 75 percent for plants commissioned between 2009 and 2017, mostly private.[2] CIL's near-monopoly remained, and only ten to twenty percent of its output was sold through e-auctions. No independent regulator was created to enforce the supply guarantees written into coal contracts or to adjudicate disputes between CIL and its customers. The Coal Regulatory Authority Bill, introduced in Parliament in December 2013, would have created such a regulator (PRS Legislative Research, 2013). The authority was to specify the principles and methodologies for determining the price of coal for all producers; to adjudicate disputes between coal companies and their customers over the grading, quality, pricing and supply of coal, and to advise the government on promoting competition, efficiency and investment in the coal industry. The bill lapsed with the dissolution of the Lok Sabha in 2014, and no statutory coal

[1] The revised Presidential directive is reproduced in Annexure VI of the CIL Annual Report, 2013–14: https://d3u7ubx0okog7j.cloudfront.net/documents/2013-14_8w1zW4u.pdf.

[2] Ministry of Coal notification of 26 July 2013, https://coal.nic.in/sites/default/files/2020-01/260713_0.pdf.

regulator has been established since.

Private generators thus entered electricity under credible contracts for their output and expropriation risks from political opportunism for their input. The asymmetry matters for the distortion. Once the price of a generator's output is fixed by its bid and protected by a regulator, the input market is the remaining channel through which the state can reallocate the generator's quasi-rents. The distortion therefore arises from the gap between commitment in the electricity market and the market for coal.

The history of private access to coal illustrates these expropriation risks. Very early in the period, there was a clear recognition that CIL, given its capacity and governance structure would be unable to meet demand from a burgeoning electricity sector. The government therefore had two alternatives to the status quo. The hypothetical first-best, deregulate coal pricing and allow competitive entry into mining, ran against a constraint that the political process treated as binding: CIL had to remain with its market presence intact, and any reform that conflicted with the "public-interest" mandate under which CIL operated was a non-starter. The second-best alternative was probably auctioning of coal-bearing reserves to private end-users, with auction rules designed such that the resulting economic outcomes approximate the social optimum within this second-best. However, the government at the time argued that the design, legislative approval and implementation of such an auction system would conflict with the urgency of meeting coal demand from a rapidly expanding power sector and the inability of CIL to meet this demand (Government of India, 2012).

The chosen instrument, instead, was administrative allocation of captive coal blocks to a highly selected subset of private generators, with royalties set at very low rates such that the overwhelming share of the resource rent stayed in the hands of the allottees. The terms were attractive: private companies that received these allocations had announced (and subsequently commissioned) nearly 17 GW of generation capacity by 2014. Because the leases were granted on a long-term basis, the expected return on investment was sufficient to outweigh the latent risk of future expropriation, and a substantial wave of private capacity entered the system on the strength of this implicit bargain.

The bargain rested on no commitment device and did not last. The Comptroller and Auditor General's assessment of the allocations in 2012 identified large losses to the exchequer (Comptroller and Auditor General of India, 2012), and in 2014 the Supreme

Court cancelled 204 of the 214 blocks allotted. The Coal Mines (Special Provisions) Act of 2015 then mandated competitive auctions for private allocations at high reserve prices, while preserving administrative allotment for public-sector entities at nominal prices. Understandably, there was very limited participation from the private sector at these auctions.

Private generators that tried to escape the domestic system fared no better. Anticipating supply risks, some built plants designed to run on imported coal, roughly 12 GW of capacity. The investments remained extremely profitable until 2014, when the Indonesian government (the primary source of coal imports for these generators) mandated that coal exports be priced at international benchmark rates rather than previously negotiated long-term contract prices, substantially increasing fuel costs for these import-based plants (Taneja, 2013). Their power purchase agreements, protected by the electricity regulators, fixed the price of their output; nothing protected the price of their input. The compensatory tariffs that regulators granted to two of these plants were struck down by the Supreme Court in 2017 (Supreme Court of India, 2017). Without a regulator for coal, private generators had no forum in which their input-cost risk could be shared, while public generators were protected by the allocation rules.

3.4 Common agency and governance by prohibition

The commitment argument leaves open the reason the state built commitment devices in electricity but not in coal. Specifically, why is it so hard to restructure CIL as a purely commercial enterprise, allowing price deregulation and contracting based on who delivers the highest social value rather than ownership class? The answer lies in the structure of CIL's principals. Delegating the coal market to an independent regulator, or exposing CIL to competition, would have required each principal to give up the lever it uses on CIL, and the principals of CIL are many and their claims conflict.

The theory. When an agent serves several principals whose interests partly conflict, and each can influence the agent's actions, the situation is one of common agency (Bernheim and Whinston, 1986). Each principal offers the agent its own incentive scheme, taking the schemes of the others as given, and the outcome is the Nash equilibrium of this game among the principals. Dixit (1996) shows, building on Holmstrom and Milgrom (1991), that the incentives the agent faces in this equilibrium are

much weaker than those a single principal would offer. The reason is free-riding. A principal that rewards the agent for an outcome it values pays the full cost of the reward, but part of the agent's response leaks to the other principals: some of them value the same outcome and gain without paying, and others oppose it and offset the reward with penalties of their own. Each principal anticipates the leakage and offers weaker incentives, or none, and the other principals do the same. In the symmetric case, the power of equilibrium incentives falls roughly in proportion to the number of principals, and it falls further when their interests are opposed (Dixit, 1996). With weak incentives, rewards for outcomes are a poor instrument for directing the agent. Holmstrom and Milgrom (1991) show that when incentives are weak, the principals do better by restricting the set of actions the agent may take than by rewarding the actions they value. Wilson (1989) describes the result in government bureaucracies: agencies are governed by constraints rather than incentives, and they can stop things from happening but find it difficult to get anything positive done.

Application to CIL. The two institutional features described above mean that CIL operates simultaneously under the influence of several principals, all of whom have some power to influence its actions. Their interests in CIL's (through Ministry of Coal) output are at least partly in conflict, and CIL's actions taken on behalf of different principals are partial substitutes and difficult to reconcile.

In a setting with a single principal, the second-best solution might be to design an incentive scheme that rewards CIL for efficiency, output expansion, or setting the price of coal closer to its marginal social value. But with multiple principals, as in the status quo, the Nash equilibrium incentive scheme would have a fraction of the power of such a second-best scheme. To illustrate, suppose the fiscal authority were to attach a high-powered incentive to CIL's profitability, rewarding it for setting competitive prices and increasing output. Every additional dollar of revenue would translate, at least in the short term, into a higher cost of electricity for the agricultural principal whose cheap-power claim is met through inexpensive thermal generation. That principal, exerting influence through state-level political pressure and through directives issued via the Ministry of Coal, would respond by imposing penalties on CIL or extracting offsetting transfers, clawing back the gain the fiscal authority had purchased. Anticipating the leakage, and the fact that providing incentives costs effort and money, the fiscal authority would respond to this leakage or free-rider problem by setting a significantly weaker or no incentive scheme in the first place, and the

same calculation runs in reverse for any incentive the agricultural principal might place. A reward from the coal-bearing states for higher output would be offset in the same way by the fiscal authority, for whom output that does not earn its cost reduces dividends.

Governance by prohibition. CIL's governance has the form the theory predicts: its principals constrain it rather than reward it. CIL is told what it may not charge (prices on the bulk of its output are set by notification rather than negotiation); to whom it must supply, and in what proportion (supply guarantees are pre-specified by class of generator and by sectoral priority); under what terms it may contract (fuel supply agreements are written under conditions CIL does not set); and what it may not exit (the workforce, mine portfolio and capital structure can be restructured only through legislation, not by commercial decision). The differentiated treatment of generators derives itself from this set of prohibitions: the trigger levels of 90 percent for older, mostly public plants and 75 percent for newer, mostly private ones, and preferential administrative allocation for public-sector plants under the 2017 SHAKTI policy, which required private generators to bid for linkages (Ministry of Coal, 2017).

In this equilibrium, any change on any margin requires the assent of every principal that can credibly retaliate. The two arguments of this section meet here. An independent coal regulator would be a commitment device of the kind built for electricity, but its creation would remove the prohibitions through which each principal governs CIL: the notified price that serves the cheap-power constituency, the allocation priorities that serve the states that own generators, and the discretion over output that serves the coal-bearing states. No principal can delegate the coal market without the consent of the others, and each of the others loses its lever. Electricity, by contrast, was reformed when the State Electricity Boards were insolvent and the cost of the status quo fell on the states themselves (Dubash and Rajan, 2001). The equilibrium also accounts for the persistence of the allocation across governments of different parties and different commitments to markets. The Electricity Act of 2003, the New Coal Distribution Policy of 2007 and the SHAKTI policy of 2017 were adopted by governments led by different parties, and the distortion survived all three. The same governments that preserved notified coal prices liberalised electricity generation, a contrast that an explanation based on ideology cannot account for and that the framework attributes to the presence of commitment devices in one market and their absence in the other. A reform that moved CIL toward a single principal, through full commercialisation

or the privatisation of its productive units, would face the same obstacle: it requires the suppression of claims that the political process treats as binding.

4 Two Feasible Reforms

The framework of Section 3 determines the form of a feasible reform. The first best, deregulating coal prices and opening mining to competition while ignoring the principals' claims, fails the *remediableness* test: it is not implementable, because it requires the suppression of claims that the political process treats as binding. A feasible reform must leave each principal that can block it no worse off on the dimension it controls. Because the distortion lies in the price of coal at the margin (Section 2), and not in the inframarginal transfers the notified price delivers, there are two ways to do this. The first route *compensates within the prohibitions*. Every lever the principals use, the notified price, the allocation rules and the supply contracts, stays in place, and the fiscal authority pays to equalise prices at the margin. The second route *compartmentalises*. Dixit (1996) shows that when each principal is restricted to acting on the dimension that primarily concerns it, the free-riding that weakened incentives disappears: each principal internalises the full marginal value of the action it cares about. Applied here, the coal price is left to reflect scarcity, and the cheap-power objective is met with a separate instrument, a cash transfer to consumers. Each route requires a commitment: the first, that the fiscal authority continues to pay compensation to generators; the second, that the Centre continues to pay the transfer to consumers. Neither commitment requires a device outside the existing political process. On both routes, every principal that can block the reform is no worse off on the dimension it controls and keeps the lever it uses today, so a principal that reneged, for example a Centre that stopped paying the transfer, would face retaliation from the principal it harmed, through the same levers that sustain the status quo. Given that retaliation, no principal gains by deviating: the reformed arrangement is a Nash equilibrium of the game among the principals, and the opportunity for political gains that undermined promises to private investors is taken away.

This section describes one reform on each route. Both give every generator the same price of coal per kcal at the margin, and both have precedents in Indian policy. The framework predicts that the two reforms produce the same efficiency gain but distribute it differently, and the estimates in Section 6 bear this out. Both reforms yield \$4.7 billion a year, \$1.9 billion in fuel and transport savings and \$2.8 billion

in avoided CO_2 damages. Under the compensatory tariff, consumers' electricity bills fall by $7.3 billion a year and the Centre pays $4.9 billion net of CIL's gains. Under deregulation, CIL's higher revenue funds the transfer, consumers are held harmless on their bills, and the Centre gains $0.2 billion. On both routes, the residual losers are the principals whose claim rests on quantities rather than prices: the state governments whose generators lose a fifth of their output.

4.1 A compensatory tariff

The compensatory tariff leaves the structure of prohibitions intact. CIL continues to supply coal under existing supply contracts at notified prices, and each generator keeps its contracted tonnage. Any coal a generator buys beyond its contract, through e-auctions or imports, is bought at the market price, and the central government reimburses the gap between that price and the notified price. In the wholesale markets, every generator therefore offers electricity as if it paid the notified price per kcal. Any unused contracted coal is then routed to the coal markets by CIL, where it sells at the market price. The policy explicitly maintains CIL's output, a binding constraint on the political process.

Importantly, the reform does not alter the political process through which the principals in Table 1 exercise influence over coal policy. Notified prices, sectoral priorities and the terms of supply contracts are unchanged, and so are CIL's output, workforce and royalty payments. The cheap-power objective is met through subsidized coal pricing as before. The cost falls on the fiscal authority (central government), which pays the compensation but is still able to balance its budget by reducing, for example, existing welfare transfers to customers who gain from lower electricity prices.

India has used this mechanism in recent years. In May 2022, facing a shortage of domestic coal, the Ministry of Power invoked Section 11 of the Electricity Act to direct all plants designed to run on imported coal to operate at full capacity. Because their long-term contracts did not allow high import prices to be passed through, a committee of the Ministry of Power, the Central Electricity Authority and the Central Electricity Regulatory Commission set benchmark rates that covered the plants' fuel costs, and the direction was renewed for the summer of 2023.[3] Earlier, between 2014 and 2017, the Central Electricity Regulatory Commission granted compensatory

[3]Ministry of Power directions under Section 11 of the Electricity Act, 2003, dated 5 May 2022 and renewed for 16 March to 15 June 2023; see also CERC Order in Petition No. 128/MP/2022, https://cercind.gov.in/2022/orders/128-MP-2022.pdf.

tariffs to the two Mundra plants after Indonesia's export pricing rule raised their fuel costs (Central Electricity Regulatory Commission, 2014a,b), until the Supreme Court set them aside in 2017 (Supreme Court of India, 2017).

4.2 Deregulation with a direct benefit transfer

The second reform compartmentalises. Coal prices are deregulated, so that CIL sells all of its coal at a price that reflects scarcity, here the import-parity price. The reform does not open mining to private entry on a scale that would displace CIL. CIL remains the dominant producer, and its output, workforce and royalty payments are unchanged, so the development mandate on which the coal-bearing states' claim rests is preserved. The cheap-power objective is met with a separate instrument: a direct benefit transfer to electricity consumers, of the kind proposed in the Draft Electricity (Amendment) Bill of 2020 (Ministry of Power, 2020). State commissions determine tariffs without reference to the implicit coal subsidy, and the government pays each state the increase in its electricity bill through dedicated accounts released to utilities against power supplied. The transfer holds both consumer principals harmless: neither the cheap-power constituency nor electricity consumers at large pay more for electricity than in the status quo. The fiscal authority is no worse off either. It funds the transfer from the windfall in CIL's revenue, since CIL now sells its contracted coal at the market price, and that windfall covers the transfer almost exactly, so the reform is at least budget neutral for the Centre (Section 6). Each principal is thereby restricted to the dimension that primarily concerns it: the coal-bearing states to CIL's output and employment, the consumer principals to the transfer, and the fiscal authority and the coal market to the price of coal. No principal can claw back an incentive placed on another principal's dimension, so the free-riding that weakened CIL's incentives disappears (Dixit, 1996), and at a single market price, coal reaches the generators that value it most, regardless of their ownership.

India has moved from subsidised prices to cash transfers and market prices in other energy markets. Since 2014, cooking-gas subsidies have been paid as direct transfers to households' bank accounts under the PAHAL scheme, and consumers buy their cylinders at the market price (Press Information Bureau, 2015). Diesel prices have been market determined since October 2014 (Press Information Bureau, 2014).

Implementing either reform requires administrative work. Under deregulation, existing supply contracts would need to be revised to reflect market prices, and long-term

power purchase agreements would pass the higher costs through to utilities, for which the Electricity Act of 2003 already provides the legal apparatus. The transfer would require payment infrastructure at scale. Under the compensatory tariff, the government must verify the coal each generator buys beyond its contract, which the Section 11 directions already required.

5 Evaluating the Reforms

5.1 The model

I evaluate both reforms with the empirical model of India's daily electricity supply developed in Deshwal (2026). The model covers more than 200 coal and lignite generators and every state's daily demand. Generators with long-term power purchase agreements (PPAs) supply only the states with which they are contracted, are paid their variable cost, and declare themselves available only when that cost is below their contracted price; generators without PPAs sell in a day-ahead market that meets the residual demand. Generators' marginal costs are constructed from their coal sources, coal prices and quality, transport costs and heat rates. Each day, given the generators that declare themselves available, a system operator chooses each generator's supply to each state to minimize the total variable cost of meeting every state's demand:

$$\min_{q_{ist}\geq 0} \sum_{i,s} c_{it}\, q_{ist} \quad \text{s.t.} \quad \sum_{i} q_{ist} = D_{st}\ \forall s, \qquad \sum_{s} q_{ist} \leq \bar{q}_{it}\ \forall i, \qquad q_{ist} \leq \bar{q}_{ist}^{PPA}\ \forall i, s, \tag{1}$$

where c_{it} is generator i's marginal cost on day t, D_{st} is state s's demand, $\bar{q}_{it}$ is the generator's available capacity and $\bar{q}_{ist}^{PPA}$ its contracted capacity for state s, which equals its available capacity for generators without PPAs. I solve this linear program numerically for every day of the sample. The model contains no parameters fitted to generation outcomes, yet it explains 85 percent of the daily variation in generators' output. I report annual averages over July 2021 to December 2023.

I compare three regimes. The *status quo* simulates supply at generators' observed costs. The *compensatory tariff* simulates supply when every generator bids its cost with all of its coal priced at the notified price per kcal, which is the counterfactual of the companion paper. Under *deregulation*, every tonne of coal is priced at the import-parity price per kcal, scaled to the generator's coal quality and including freight, cess and import duty. Both reforms give every generator the same price of

coal per kcal at the margin, so I hold generation under deregulation at its level under the compensatory tariff and recompute only costs and prices.[4]

5.2 Coal flows

Let q_i be generator i's output under the reform and q_i^0 under the status quo, t_i the tonnes of coal it burns, k_i the tonnes it receives under its supply contract in the status quo, and p_i^n and p_i^m the notified and market prices per tonne of the coal it burns. CIL's output is fixed: it supplies k_i under each contract, and contracted coal a generator no longer burns reaches other generators at the market price, so the change in imports equals the change in coal burned.

Under the compensatory tariff, generator i pays p_i^n on $\min(k_i, t_i)$ tonnes and p_i^m on the remainder, and the government reimburses $(p_i^m - p_i^n)$ on the remainder. CIL earns $(p_i^m - p_i^n)$ more than in the status quo on the $\max(k_i - t_i, 0)$ tonnes it resells to other generators. Under deregulation, generator i pays p_i^m on all t_i tonnes, there is no compensation, and CIL earns $(p_i^m - p_i^n)$ more on all k_i tonnes. Lignite generators are supplied from their own mines; I hold their costs at observed values in all regimes, so every lignite term is identical across the two reforms.

5.3 Payments and prices

PPA-based generation is paid the generator's average cost, net of any compensation; generation for the day ahead market is paid the day's market price. In the model, the market price on day t is the highest cost among dispatched coal and lignite generators,

$$P_t = \max_{i:\, q_{it}>0} c_{it}, \tag{2}$$

which reproduces the model's price on every day under both the status quo and the compensatory tariff (Appendix A). Under deregulation, I apply equation (2) to the deregulated costs of coal generators only, because lignite costs cannot be repriced at import parity; on every day of the sample, the highest-cost dispatched generator under deregulation is a coal generator in any case. Each generator's payments are assigned to the states that buy its output, in proportion to its contracted and uncontracted

[4] Coal costs scale with the coal price while freight and other costs do not, so the merit order under a uniform market price can differ slightly from that under a uniform notified price, and generators' availability decisions can differ when their costs cross their contracted prices. The effect of these differences on generation is negligible.

supply to each state. A state's electricity bill B_s is the sum of these payments.

5.4 Principals

I group the gains and costs by the principals of Table 1:

- *Consumers*, by the state that buys the electricity: the change in their electricity bill, plus any direct benefit transfer, plus avoided CO_2 damages, valued at \$190 per tonne.
- *The Centre and CIL*, treated as one principal: compensation paid, CIL's revenue above the notified price, the direct benefit transfer, and cess and import duty.[5]
- *Coal-bearing states*: because coal output does not change, I hold royalty and other levies at their status-quo values.[6]

Under deregulation, the direct benefit transfer holds each state's consumers harmless: state s receives $\text{DBT}_s = \max(\Delta B_s, 0)$, the increase in its electricity bill. Excluding avoided CO_2 damages, the sum of the changes across principals equals the saving in fuel, freight and secondary fuel, valued at the international price of coal per kcal. Because CO_2 is a global pollutant, state-level results allocate avoided CO_2 damages to the states that buy the electricity in proportion to their purchases.[7]

6 Distribution of Impacts

Both reforms give every generator the same price of coal per kcal at the margin, so they produce the same reallocation of generation, from state-owned and central-sector generators to private generators without supply contracts. The reallocation saves \$1.9 billion a year in fuel and transport costs and reduces CO_2 emissions by 14.6 million tonnes, worth \$2.8 billion at \$190 per tonne: \$4.7 billion a year in all. The two reforms distribute this gain differently, across the principals of Table 1 and across states.

[5]The Government of India held 63.13 percent of CIL's equity as of December 2023 (Reuters, 2026), so part of CIL's gain accrues to minority shareholders.

[6]Royalty is levied at 14 percent of the coal price, so under deregulation the coal-bearing states would gain if royalty were levied on the deregulated price. I assign the whole gain to CIL and the Centre.

[7]The damages from a global pollutant need not be distributed in proportion to consumption. Extreme temperatures may be concentrated in some regions and cyclones in some coastal areas, so that the damages avoided by a reduction in emissions differ across states. Allocating them in proportion to purchases is a simplifying assumption.

6.1 Distribution across principals

Table 3 reports the annual change in welfare for the consumer principals and for the Centre and CIL, and Table 2 reports the change in generation by group of generators.

Consumers. The cheap-power constituency and electricity consumers at large buy electricity through two markets: long-term contracts, which carry about 80 percent of coal and lignite generation, and the short-term market, which carries the rest. Under the compensatory tariff, the average price of contracted electricity falls from INR 3.30 to INR 2.94 per kWh, because generators pass the compensated cost of their marginal coal through their variable charges, and the average short-term price falls from INR 5.26 to INR 4.09 per kWh, because the generators that set it now bid at the notified price of coal. Consumers' bills fall by $7.3 billion a year, and with $2.8 billion of avoided CO_2 damages, consumers gain $10.0 billion. Under deregulation, the average contract price rises to INR 6.50 per kWh and the average short-term price to INR 8.36 per kWh, and bills rise by $47.2 billion a year. The direct benefit transfer pays each state the increase in its bill, so both consumer principals are held harmless, and they gain the $2.8 billion of avoided CO_2 damages.

The fiscal authority and CIL. Under the compensatory tariff, the Centre pays $10.4 billion a year in compensation and recovers $5.2 billion through CIL's sale, at the market price, of contracted coal that state-owned generators no longer burn. Its net cost of $4.9 billion is less than the $7.3 billion fall in consumers' bills, so the Centre can balance its budget by reducing other transfers to the consumers who gain. Under deregulation, CIL's revenue on its contracted coal rises by $47.1 billion, which funds the $47.2 billion transfer almost exactly; with cess and import duty, the Centre gains $0.2 billion. The reform is at least budget neutral.

Coal-bearing states. Under both reforms, CIL's output, workforce and royalty payments are unchanged, so the claims of the coal-bearing states are unaffected.

States that own generators. Their generators lose output under both reforms. State-owned generation falls by 72 TWh a year, 21 percent of its status-quo level. Valued at the average status-quo price of coal and lignite generation, INR 3.68 per kWh, the lost output is worth $3.3 billion a year. The loss is one of utilisation, and of the public employment and political value that state governments attach to their own generators running. Section 6.2 measures it state by state.

Generating companies. Central-sector generation falls by 18 TWh a year, 4 percent,

and private generation rises by 90 TWh, 24 percent. Among private generators, those with supply contracts keep their contracted coal at notified prices under the compensatory tariff but lose 14 TWh of generation to more efficient competitors. The gain goes to private generators without supply contracts, whose generation nearly doubles, rising by 104 TWh a year, output worth $4.8 billion at the average status-quo price.

Table 2. Generation by generator group under both reforms

		Generation (TWh/yr)			Value of change
Generators	Stations	Status quo	Change	%	(bn USD/yr)
Central	46	461.8	-18.2	-3.9	-0.84
State	58	341.0	-71.9	-21.1	-3.30
Private					
With supply contracts	42	265.0	-14.1	-5.3	-0.65
Without supply contracts	26	106.4	104.1	97.8	4.78

Generation is the same under both reforms. Private generators with supply contracts: stations that receive at least 70 percent of their coal from CIL, SCCL or captive mines in the status quo, weighted by generation. Stations include lignite generators. Value of change: change in generation at the average status-quo price of coal and lignite generation, INR 3.68 per kWh, which prices contracted generation at its variable charge and short-term generation at the short-term price. Annual averages over July 2021–December 2023, at INR 80 per USD.

Table 3. Annual gains and costs by principal (billion USD)

	Compensatory tariff	Deregulation + DBT
Consumers		
Change in electricity bills	7.27	-47.22
Direct benefit transfer	–	47.22
Avoided CO_2 damages	2.78	2.78
Net	10.05	2.78
Centre and CIL		
Compensation paid	-10.40	–
CIL revenue above notified price	5.19	47.06
Direct benefit transfer paid	–	-47.22
Cess, import duty and lignite	0.33	0.33
Net	-4.88	0.17
Total	**4.70**	**4.70**

Changes relative to the status quo. Compensatory tariff: the government reimburses generators the gap between the market and notified price on coal bought beyond their contracts. Deregulation: all coal is priced at the import-parity price, and a direct benefit transfer (DBT) pays each state the increase in its electricity bill. Generation is the same under both reforms. Consumers: change in electricity bills of the buying states, the DBT, and avoided CO_2 damages valued at \$190 per tonne. Centre and CIL: compensation, CIL's revenue above the notified price, the DBT, cess, import duty and lignite. The total equals fuel and transport savings plus avoided CO_2 damages. Annual averages over July 2021–December 2023, at INR 80 per USD.

6.2 Distribution across states

Among the various principals, the state governments that own generators are the only losers. I therefore focus on the 14 states with state-owned generators and compare two measures: the gain to their electricity consumers, and the relative economic importance of the lost output of their own generators. Table 4 reports both, and Figure 2 visualizes them on a map.

I measure consumers' gain by the change in the average price the state pays for electricity from coal and lignite generators, including contracted purchases at the PPA price and short-term purchases at the short-term price. Avoided CO_2 damages add INR 0.19 per kWh in every state under both reforms. Under the compensatory tariff, the average price falls in every state that buys at least 5 TWh a year, by between 6 and 25 percent. The fall is largest in states that buy from private plants supplied with imported or auctioned coal, whose costs fall most under the reform: Gujarat (INR 1.00 per kWh, 25 percent), Haryana (0.89, 21 percent), Punjab (0.83,

20 percent) and Tamil Nadu (0.76, 18 percent).[8]

I measure the relative economic importance of the state-owned generators' loss by the value of their lost output, the change in their generation valued at the average status-quo price of coal and lignite generation (INR 3.68 per kWh), relative to the state's gross domestic product. The measure serves as a proxy for the scale of lost economic activity: revenue and employment effects from lower utilization of generators, and the political value they carry. The relative losses are small in magnitude. The largest, in Andhra Pradesh, is worth 0.38 percent of the state's GDP; Haryana's is 0.24 percent, and the median across the 14 states is 0.12 percent.

Figure 3 compares the two measures. States whose generators lose more also tend to see larger reductions in electricity prices: the correlation between the reduction in the average price and the value of lost output is 0.49, and the rank correlation is 0.58. Gujarat, Haryana, Punjab and Tamil Nadu lose output worth 0.12 to 0.24 percent of their GDP but also gain the most per kWh in energy savings. Andhra Pradesh is the exception: its generators lose output the most, relative to its economy, and its consumers gain little.

Under deregulation, the direct benefit transfer ranges from INR 2.6 to 3.7 per kWh across states (Figure 2, panel c). It is largest in states that rely most on contracted coal, such as Chhattisgarh, West Bengal and Uttar Pradesh, whose bills rise most when that coal is priced at the market, and smallest in Gujarat, whose generators already buy much of their coal at market prices. The two reforms therefore direct resources to different states: the compensatory tariff to states that buy from private plants, deregulation to states that rely on contracted coal.

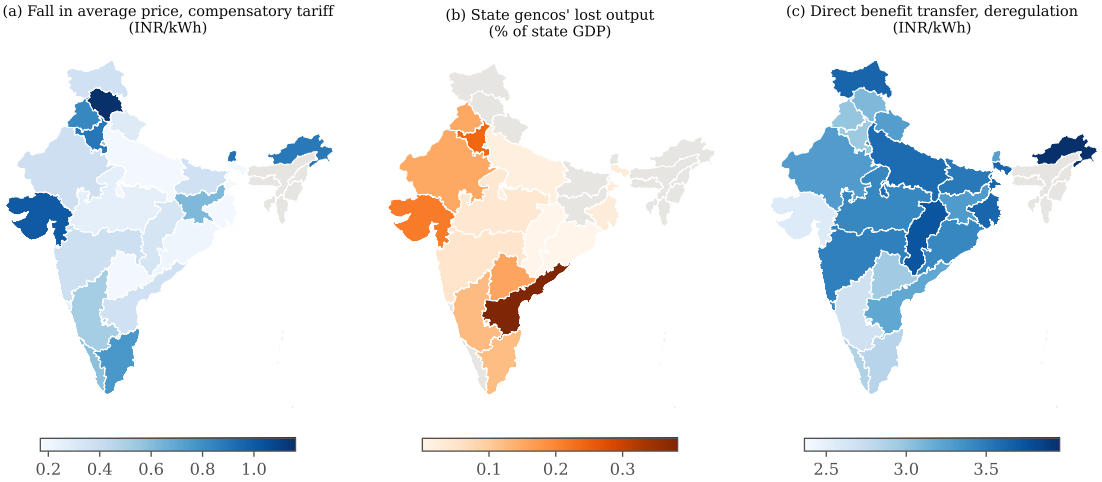
[8] States without state-owned generators gain too: Himachal Pradesh and Delhi through cheaper short-term purchases (Appendix Table A1).

Table 4. Incidence in states with state-owned generators

		Average price (INR/kWh)			State-owned generators		
State	TWh/yr	Status quo	Comp. tariff	Fall (%)	TWh/yr	mn USD/yr	% of GSDP
Andhra Pradesh	58	4.05	3.68	9.2	-13.6	-626	0.38
Haryana	58	4.20	3.31	21.2	-6.4	-295	0.24
Gujarat	105	4.08	3.08	24.5	-12.9	-592	0.21
Telangana	65	3.43	3.24	5.7	-5.6	-257	0.16
Rajasthan	67	3.54	3.16	10.8	-5.6	-256	0.15
Punjab	58	4.19	3.36	19.8	-2.8	-131	0.15
Karnataka	44	3.95	3.43	13.2	-7.9	-363	0.13
Tamil Nadu	81	4.30	3.54	17.8	-7.9	-365	0.12
Maharashtra	167	3.60	3.21	10.6	-5.2	-237	0.05
Madhya Pradesh	79	2.99	2.73	8.8	-1.4	-65	0.04
West Bengal	44	3.05	2.86	6.3	-1.2	-55	0.03
Uttar Pradesh	131	3.10	2.90	6.3	-1.3	-60	0.02
Odisha	28	2.89	2.68	7.3	0.0	0	0.00
Chhattisgarh	35	3.39	3.05	10.0	0.0	0	0.00

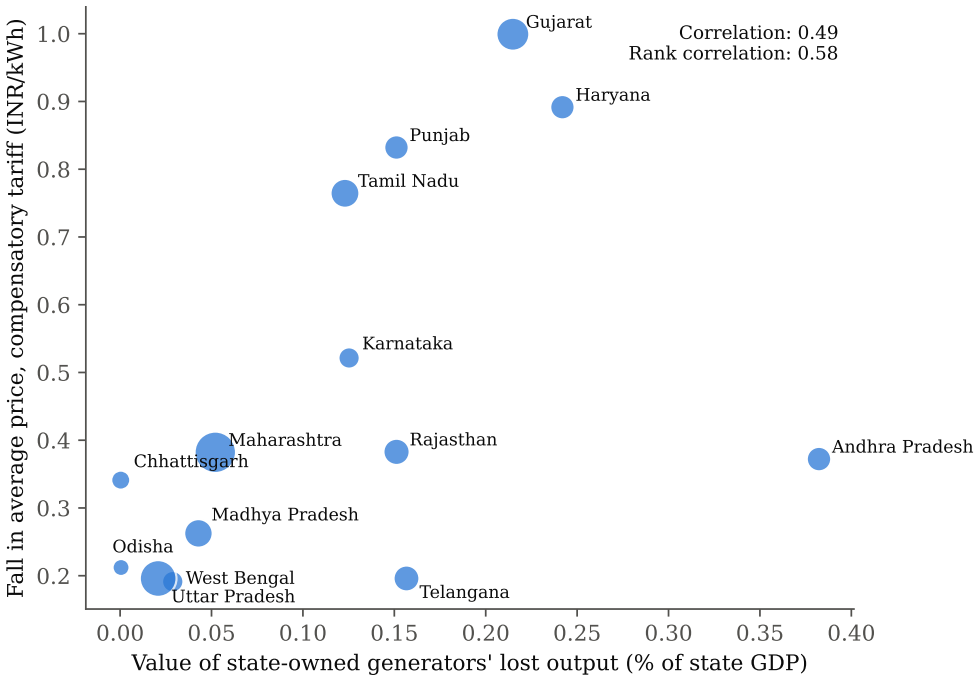
States with state-owned coal or lignite generators, ordered by the value of those generators' lost output relative to state GDP. TWh/yr: the state's purchases from coal and lignite generators in the status quo. Average price: the state's payments per kWh for electricity from coal and lignite generators in the status quo and under the compensatory tariff, and its percentage reduction. State-owned generators: change in the output of state-owned generators located in the state (TWh/yr); its value (mn USD/yr); and that value as a percentage of the state's 2022–23 gross domestic product at current prices (Reserve Bank of India, 2025). Average prices include contracted purchases at the contract price and short-term purchases at the short-term price. The value of lost output is the change in generation at the average status-quo price of coal and lignite generation, INR 3.68 per kWh; generation is the same under both reforms. Annual averages over July 2021–December 2023, at INR 80 per USD.

Figure 2. Incidence across states



(a) Fall in the state's average price of electricity from coal and lignite generators under the compensatory tariff. (b) Value of the lost output of state-owned generators located in the state, as a percentage of 2022–23 state GDP; states without state-owned generators in gray. (c) Direct benefit transfer per kWh under deregulation. States buying less than 1 TWh a year in gray.

Figure 3. Consumer gains and the value of state-owned generators' lost output



Each marker is a state with state-owned coal or lignite generators, with area proportional to its purchases. Horizontal axis: value of the lost output of state-owned generators located in the state, at the average status-quo price of coal and lignite generation, as a percentage of 2022–23 state GDP. Vertical axis: fall in the state's average price of electricity from coal and lignite generators under the compensatory tariff.

6.3 Implications for the remediableness test

The distribution of impacts shown in the previous section identifies the principals who might use the existing political process to block the reforms. The coal-bearing states are unaffected, the consumer principals gain or the impact kept unchanged, and the fiscal authority is at least as well off under deregulation and can balance its budget under the compensatory tariff. The state governments that own generators are the only group that might block coal reforms by bargaining with the Centre and by their support for coalitions at the Centre.

A state government that considers obstructing a reform weighs two groups. Its consumers gain, through lower electricity prices under the compensatory tariff and through avoided CO_2 damages under both reforms. These consumers may reward the state government through voting in the state elections, held every five years. On the other hand, its generators lose output, and with it the employment and business that state-owned generation sustains. Employees of state-owned enterprises and the firms

that supply them are potentially organised and politically valuable: politicians use state-owned firms to deliver excess employment to their supporters, so that privatisation and restructuring are politically costly (Shleifer and Vishny, 1994). Following Grossman and Helpman (1994), the state government maximizes a weighted sum of the welfare of these groups, and an organised group can carry a larger weight per dollar than diffuse consumers.

Table 4 and Figure 3 bound the political weight at which a state government would obstruct the compensatory tariff. In 13 of the 14 states with state-owned generators, consumers' gain, including avoided CO_2 damages, exceeds the value of the lost output of the state's own generators, and in 11 of them it exceeds it by more than half. A state government obstructs the reform only if it values a dollar of output of its generators more than a dollar of its consumers' gains by more than this ratio: by a factor of at least 1.9 in Rajasthan, 2.6 in Gujarat, Haryana and Tamil Nadu, and 5 or more in Maharashtra, Punjab, Madhya Pradesh and Uttar Pradesh. Andhra Pradesh is the exception: its consumers gain less than the value of its generators' lost output, and it is the state whose consent would be hardest to secure. Because the value of lost output measures gross activity, and the income lost by employees and suppliers is a fraction of it, these factors understate the weight required. Under deregulation, the counterweight is smaller, because the transfer holds consumers harmless rather than passing the fall in costs through to them: consumers gain the avoided CO_2 damages, which exceed the value of lost output in 7 of the 14 states.

The reforms therefore pass the remediableness test with a qualification on the objective of the state governments that own generators. Its form determines the incentive of these governments to obstruct a reform and, given coalition politics at the Centre, the chance that obstruction succeeds. A state government with moderate to high weight on its consumers has little incentive to block either reform, which leaves most of these states better off on balance. A state government with a low weight on consumers relative to the employees and business of its generators may obstruct either reform, Andhra Pradesh first among them. Even then, obstruction by some states need not block a reform. Coal allocation is decided by the Centre, and a state's main levers over it are its support for the governing coalition and its bargaining with the Centre. The Centre does not necessarily depend on the support of every state that owns generators, and many of these states may not be pivotal to the governing coalition at any given time. Even if half of them objected, support from the remain-

ing states, and from states without state-owned generators whose consumers benefit, could be enough for the reform to proceed. The evidence presented thus far does not establish that this principal would block either reform. I'd argue that its opposition is a minor concern, with a counterweight in the consumers who benefit from them and exercise voting in elections to express their support.

7 Conclusion

This paper applies the transaction-cost politics framework to the policies that govern the allocation and pricing of coal for electricity generators in India. These policies give the cleanest and most efficient generators the highest price of coal at the margin. This distortion imposes an efficiency cost of $4.7 billion a year. I show that the distortion persists for two reasons. First, the Indian state built commitment devices for private investors in electricity, through independent regulators and competitively bid contracts, but not in coal, where promises of coal supply are susceptible to political opportunism. Second, CIL is a common agent. It answers to several principals with conflicting claims, each of which offers it weak incentives, so that CIL is governed by prohibitions and no principal can delegate the coal market to an independent authority without the consent of the others.

The argument explains the failure of more efficient alternatives to the current allocation. Deregulating coal prices and opening mining to competition would remove the prohibitions through which each principal governs CIL: the notified price that serves the cheap-power constituency, the allocation priorities that serve the states that own generators, and the control of output and employment that serves the coal-bearing states. The alternatives that the political process did allow lacked a commitment device, and consequently, later directives and court orders reversed or revised them. The argument also accounts for the persistence of the allocation across governments of different parties: the same governments that preserved notified coal prices liberalised electricity generation, a market with commitment devices. A feasible reform must therefore leave each principal that can block it no worse off on the dimension that it controls.

Two such reforms are available for India's coal. A compensatory tariff keeps notified prices and supply contracts in place, and the Centre pays private generators the gap between the market and notified price on the coal they buy beyond their contracts. Price deregulation with a direct benefit transfer lets the price of coal reflect scarcity

and returns the increase in electricity bills to consumers, funded by CIL's higher revenue. Both yield the same $4.7 billion a year, neither harms the coal-bearing states or consumers, the Centre can balance its budget under both, and neither needs a commitment device outside the existing political process. The state governments that own generators lose a fifth of their generation, but the loss is small relative to their economies and, in all but one state, smaller than the gain to their consumers under the compensatory tariff. The reforms therefore pass the remediableness test, unless these state governments value their own generators far above their consumers. As countries add carbon pricing to existing energy subsidies, the design of those subsidies, and the political process that sustains them, become part of feasible climate policy.

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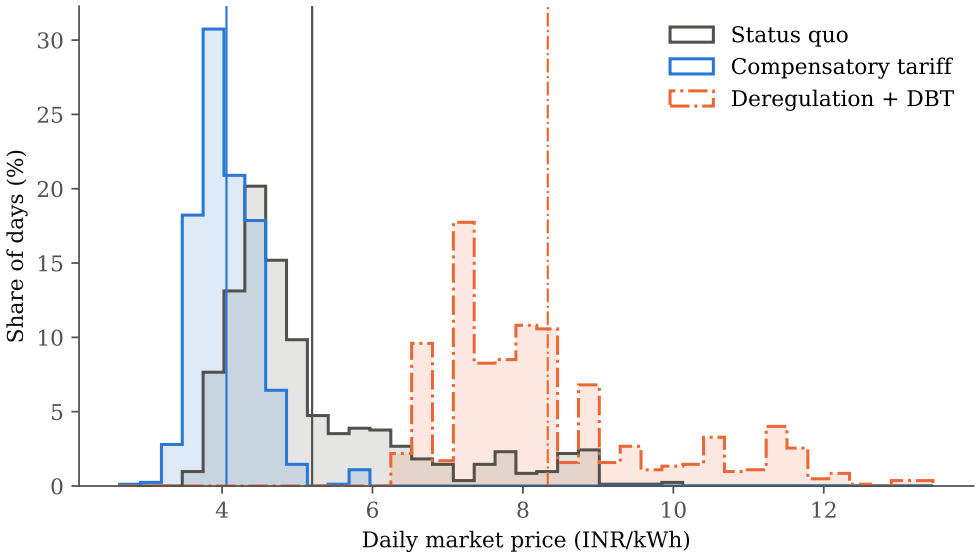
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A Market Prices under the Three Regimes

In the dispatch model, the market price on each day is the cost of the highest-cost dispatched coal or lignite generator, equation (2). Applied to the model's output, this rule reproduces the model's price on all 823 days of the sample under both the status quo and the compensatory tariff. I apply it to the deregulated costs to obtain the deregulated price. Appendix Figure A1 shows the distribution of daily prices under the three regimes.

Appendix Figure A1. Daily market prices



Distribution of daily market prices (INR/kWh) across the 823 days of the sample, July 2021–December 2023. Vertical lines mark the mean of each distribution. Under deregulation, generation is held at the compensatory-tariff dispatch and the price is the highest deregulated cost among dispatched coal and lignite generators.

B Incidence in All States

Appendix Table A1. Incidence by state

		Average price (INR/kWh)			Consumers (bn USD/yr)			State gencos
State	TWh/yr	Status quo	Comp. tariff	Dereg.	Comp. tariff bills	DBT	Avoided CO_2	(TWh/yr)
Gujarat	105	4.08	3.08	6.63	1.30	3.38	0.25	-12.9
Maharashtra	167	3.60	3.21	7.06	0.79	7.23	0.39	-5.2
Tamil Nadu	81	4.30	3.54	7.14	0.77	2.88	0.19	-7.9
Haryana	58	4.20	3.31	7.17	0.64	2.14	0.14	-6.4
Punjab	58	4.19	3.36	7.19	0.60	2.17	0.14	-2.8
Himachal Pradesh	26	5.15	3.98	8.23	0.38	1.00	0.06	0.0
Rajasthan	67	3.54	3.16	6.84	0.32	2.74	0.16	-5.6
Uttar Pradesh	131	3.10	2.90	6.69	0.32	5.87	0.31	-1.3
Andhra Pradesh	58	4.05	3.68	7.33	0.29	2.33	0.14	-13.6
Karnataka	44	3.95	3.43	6.63	0.29	1.47	0.10	-7.9
Madhya Pradesh	79	2.99	2.73	6.42	0.26	3.38	0.19	-1.4
Delhi	32	3.77	3.18	7.15	0.24	1.35	0.08	0.0
Bihar	38	3.47	3.10	6.98	0.18	1.67	0.09	0.0
Telangana	65	3.43	3.24	6.39	0.16	2.40	0.15	-5.6
Chhattisgarh	35	3.39	3.05	7.13	0.15	1.64	0.08	0.0
Kerala	18	3.62	3.04	6.49	0.13	0.63	0.04	0.0
West Bengal	44	3.05	2.86	6.68	0.11	2.01	0.10	-1.2
Sikkim	8	4.71	3.81	7.97	0.09	0.32	0.02	0.0
Odisha	28	2.89	2.68	6.32	0.07	1.20	0.07	0.0
Jharkhand	9	3.83	3.21	7.13	0.07	0.36	0.02	0.0
Jammu and Kashmir	8	3.69	3.31	7.31	0.04	0.36	0.02	0.0
Other states	17				0.07	0.72	0.04	0.0
All states	1174				7.27	47.22	2.78	-71.9

States buying at least 5 TWh a year from coal and lignite generators, ordered by consumers' gain under the compensatory tariff. TWh/yr: purchases from coal and lignite generators in the status quo. Average price: payments per kWh, including contracted purchases at generators' average cost and short-term purchases at the short-term price. Comp. tariff bills: fall in the state's electricity bill under the compensatory tariff. DBT: direct benefit transfer under deregulation, equal to the increase in the state's electricity bill. Avoided CO_2: national avoided CO_2 damages, valued at $190 per tonne, allocated in proportion to the state's purchases; the same under both reforms. State gencos: change in output of state-owned generators located in the state. Annual averages over July 2021–December 2023.